Internet Protocols EBU5403
Live Tutorial (Revision)

https://www.menti.com/alpm3k5q421q

Code: 8794 5175

Module organiser: Richard Clegg (r.clegg@qmul.ac.uk)

Jonathan Loo (j.loo@qmul.ac.uk)
Habiba Akter (h.akter@qmul.ac.uk)

	Part A	Part B	Part C	Part D
Ecommerce + Telecoms 1	Richard Clegg		Jonathan Loo	
Telecoms 2	Xianhui Che		Habiba Akter	

Reminder of lecture contents

- Lecture D3
  - VLAN
  - MPLS

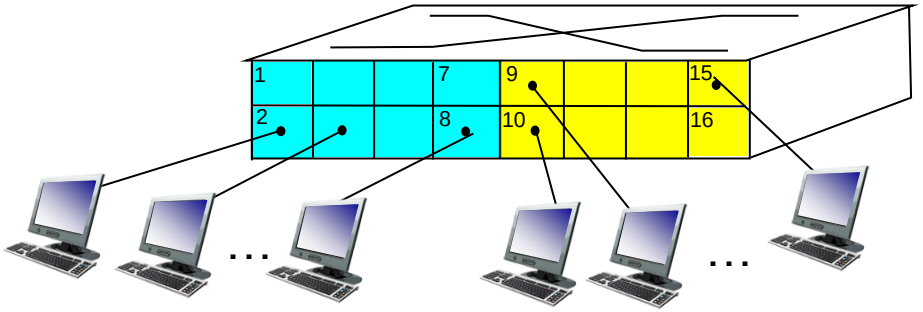


VLANs

Virtual Local Area Network

switch(es) supporting VLAN capabilities can be configured to define multiple ***virtual*** LANS over single physical LAN infrastructure.

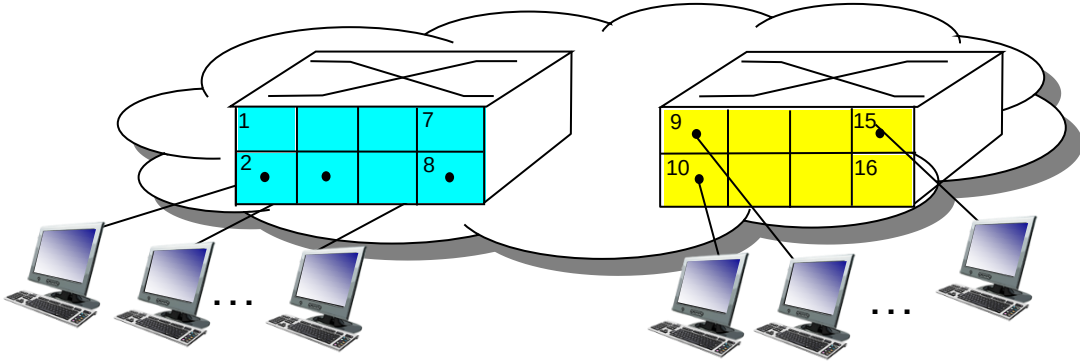
port-based VLAN: switch ports grouped (by switch management software) so that *single* physical switch ……



Electrical Engineering (VLAN ports 1-8)

Computer Science (VLAN ports 9-15)

… operates as multiple virtual switches

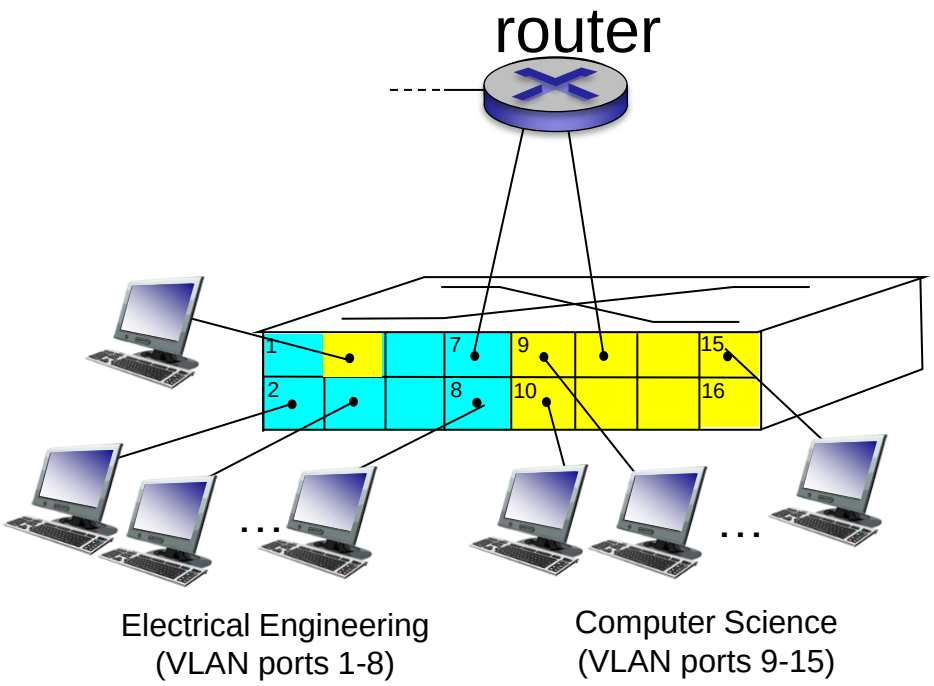


Electrical Engineering (VLAN ports 1-8)

Computer Science (VLAN ports 9-16)

Port-based VLAN

- *traffic isolation:* frames to/from ports 1-8 can *only* reach ports 1-8
  - can also define VLAN based on MAC addresses of endpoints, rather than switch port
- dynamic membership: ports can be dynamically assigned among VLANs
- forwarding between VLANS: done via routing (just as with separate switches)
  - in practice vendors sell combined switches plus routers



Question: Which of the following statements about VLANs are correct?

a) VLANs address issues such as scalability, security, and network management.
b) VLANs can partition a local network into several distinctive segments.
c) VLANs assigned users to networks based on physical topologies and distances.
d) VLANs provide the flexibility to adapt to changes in network requirements.
e) VLANs cannot control traffic patterns and react quickly to user or equipment reallocations.
f) VLANs can reduces the amount of broadcast traffic each network device and network segment



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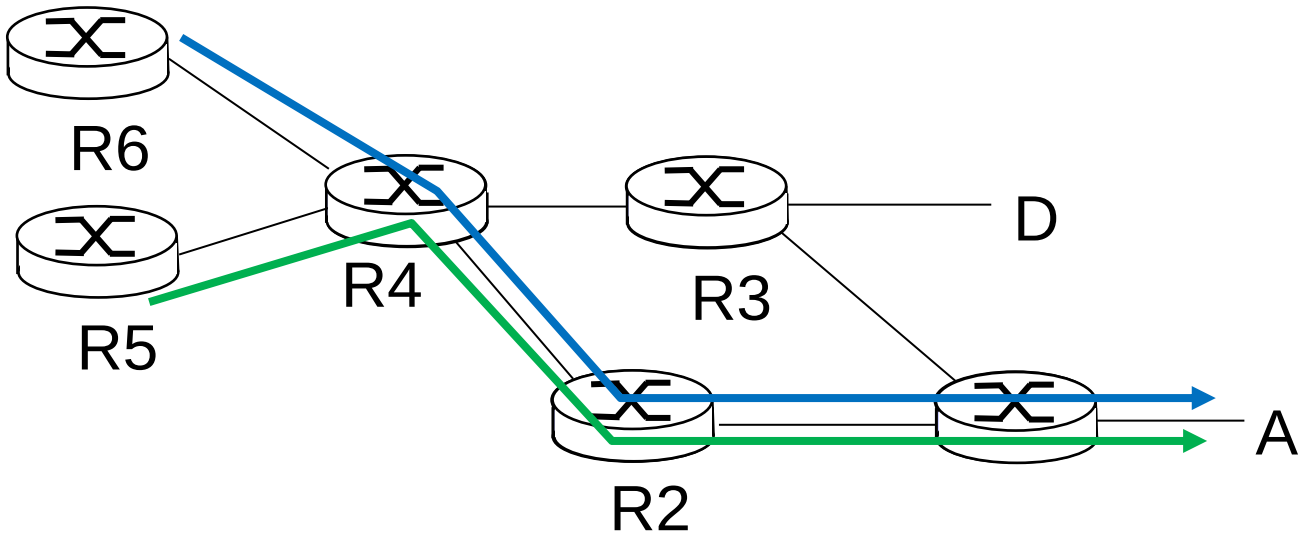
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MPLS capable routers

- a.k.a. label-switched router
- forward packets to outgoing interface based only on label value (*don't inspect IP address*)
  - MPLS forwarding table distinct from IP forwarding tables
- *flexibility:* MPLS forwarding decisions can *differ* from those of IP
  - use destination *and* source addresses to route flows to same destination differently (traffic engineering)
  - re-route flows quickly if link fails: pre-computed backup paths (useful for VoIP)

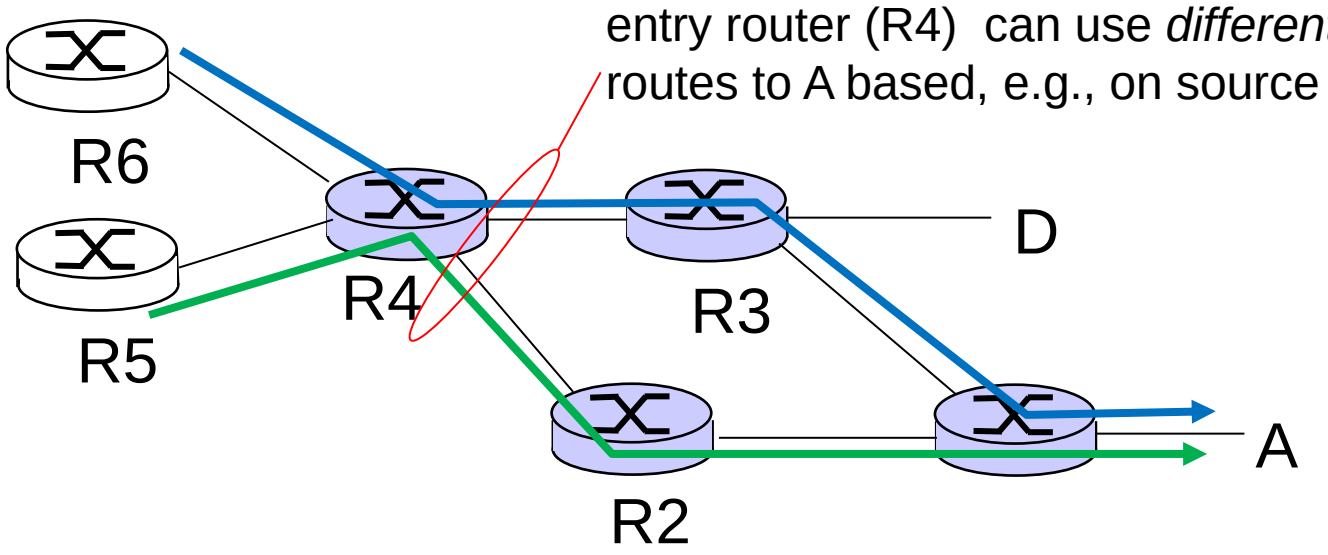
MPLS versus IP paths



- *IP routing: path to destination determined by destination address alone*


IP router

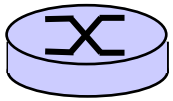
MPLS versus IP paths



- *IP routing: path to destination determined by destination address alone*
- *MPLS routing:* path to destination can be based on source *and* destination address
  - *fast reroute:* precompute backup routes in case of link failure



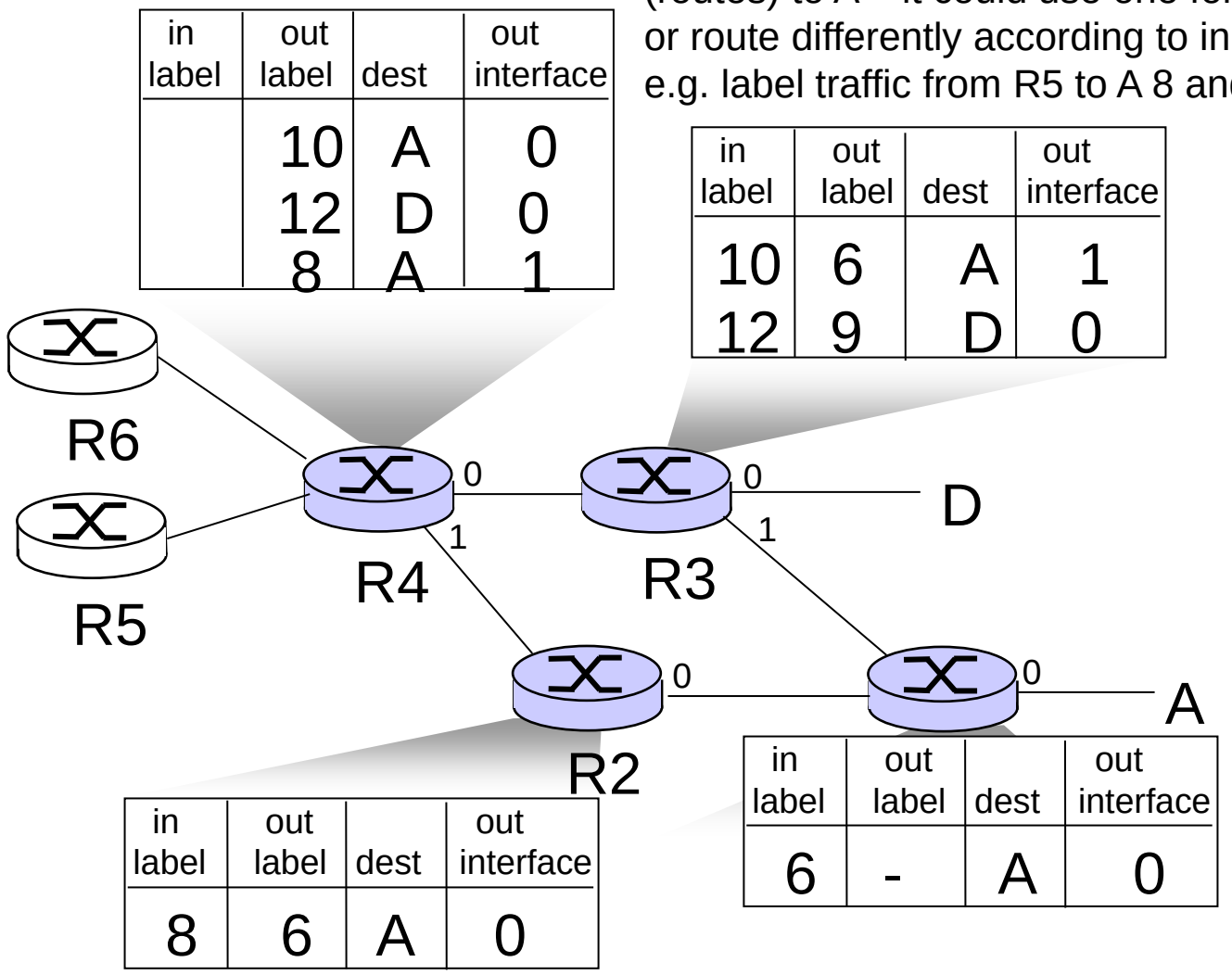
IP-only router



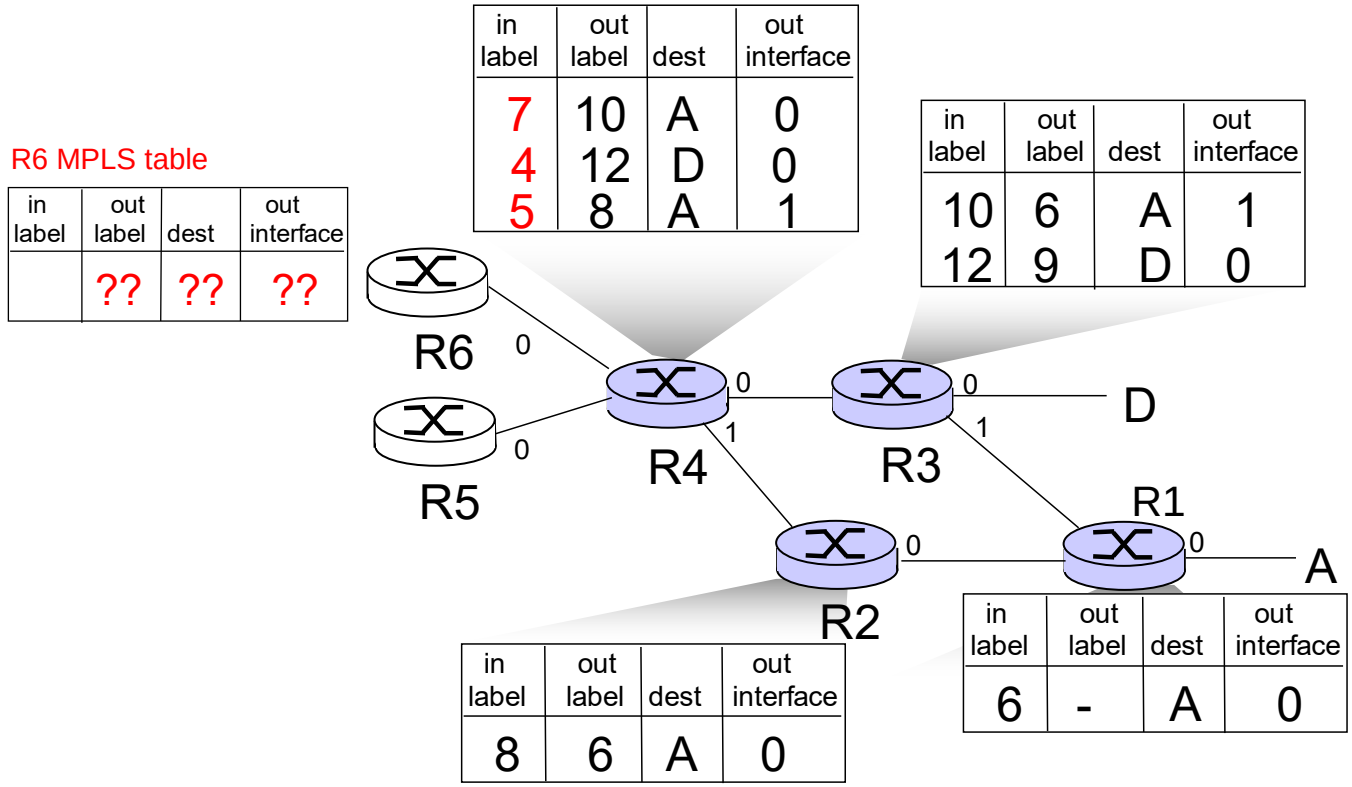
MPLS and IP router

MPLS forwarding tables

Note in this example R4 has two possible labels (routes) to A – it could use one for redundancy or route differently according to inbound router e.g. label traffic from R5 to A 8 and from R6 10



Question: Assume R6 is MLPS capable and I want to send traffic from R6 to A using the route R6-R4-R2-R1-A.



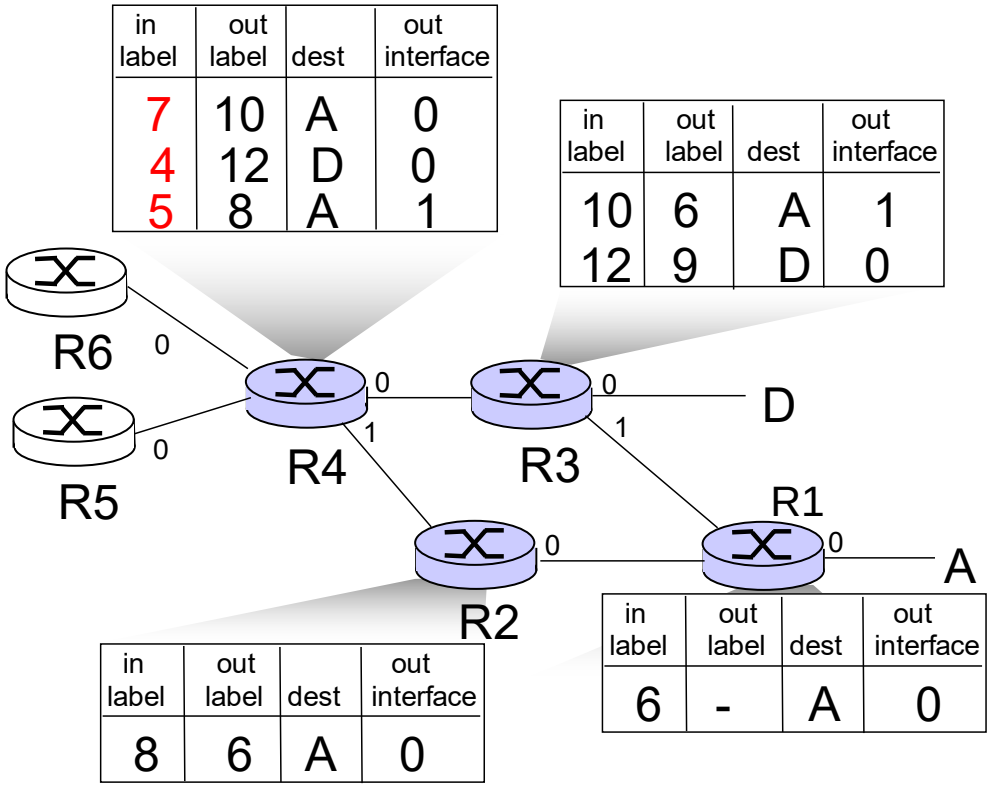
Which of the following is the correct answer?

a) 7, A, 0
b) 4, A, 0
c) 5, A, 0

Answer: Assume R6 is MLPS capable and want to switched from R6 to A using the route R6-R4-R2-R1-A.

R6 MPLS table

in label	out label	dest	out interface
	??	??	??



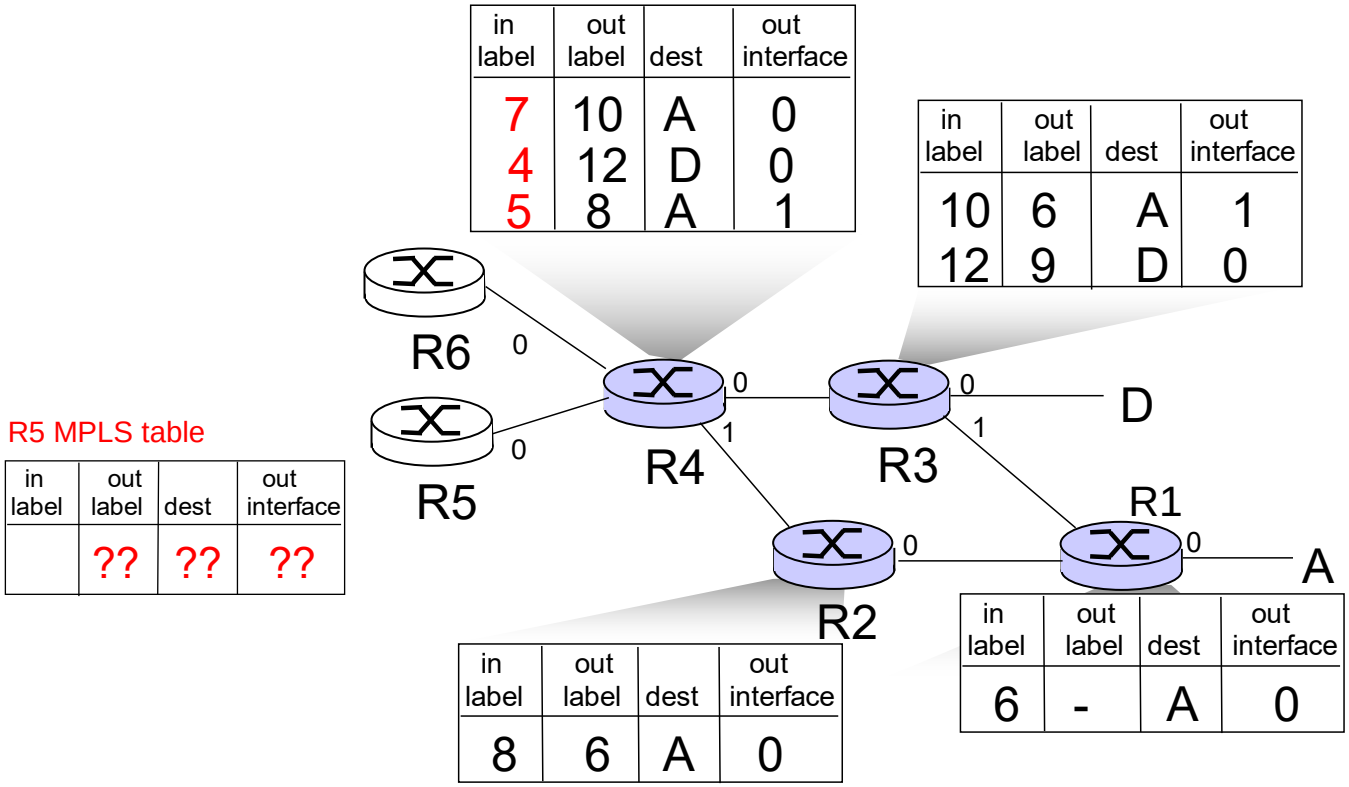
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b) 4, A, 0

c) 5, A, 0

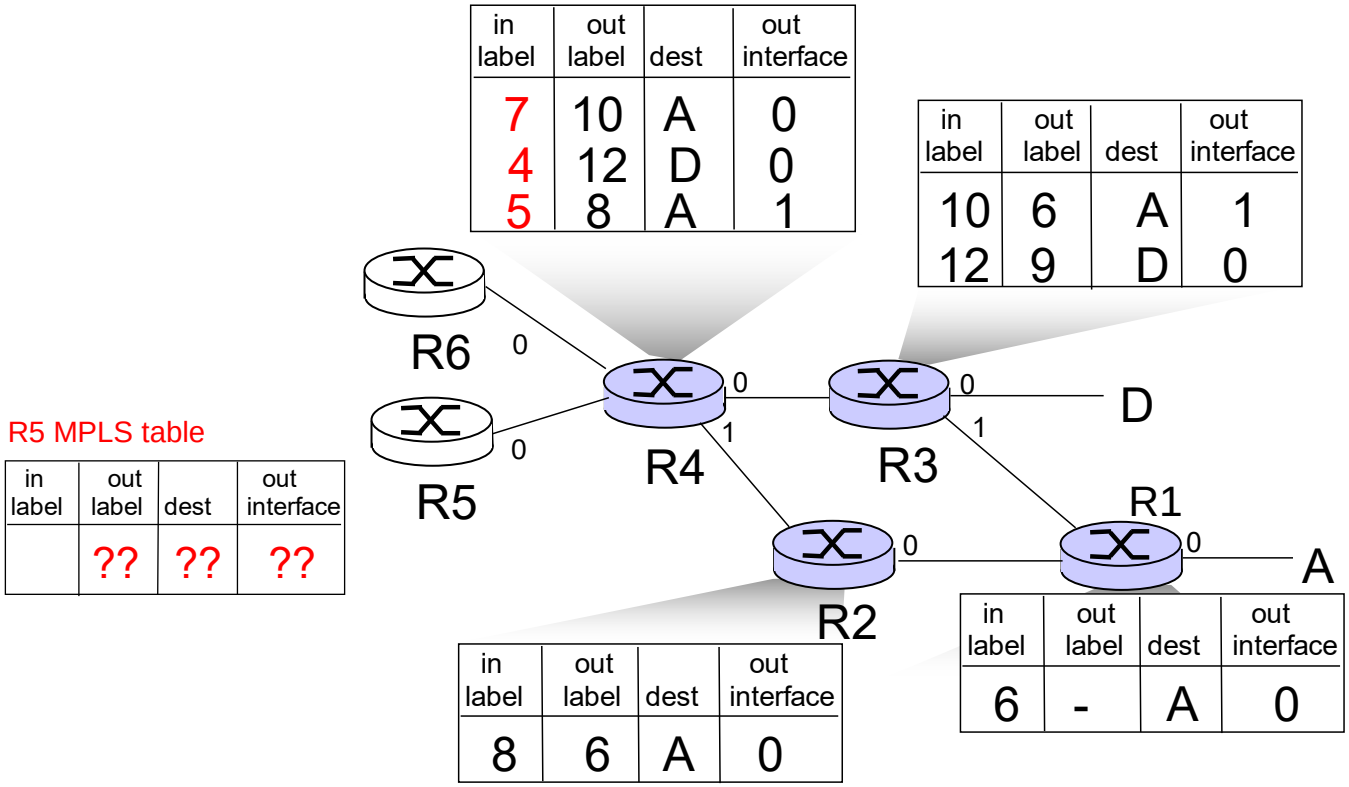
Question: Assume R5 is MLPS capable and want to switched from R5 to D using the route R5-R4-R3-D.



Which of the following is the correct answer?

a) 7, A, 0
b) 4, D, 0
c) 5, A, 0

Answer: Assume R5 is MLPS capable and want to switched from R5 to D using the route R5-R4-R3-D.



Which of the following is the correct answer?

a) 7, A, 0

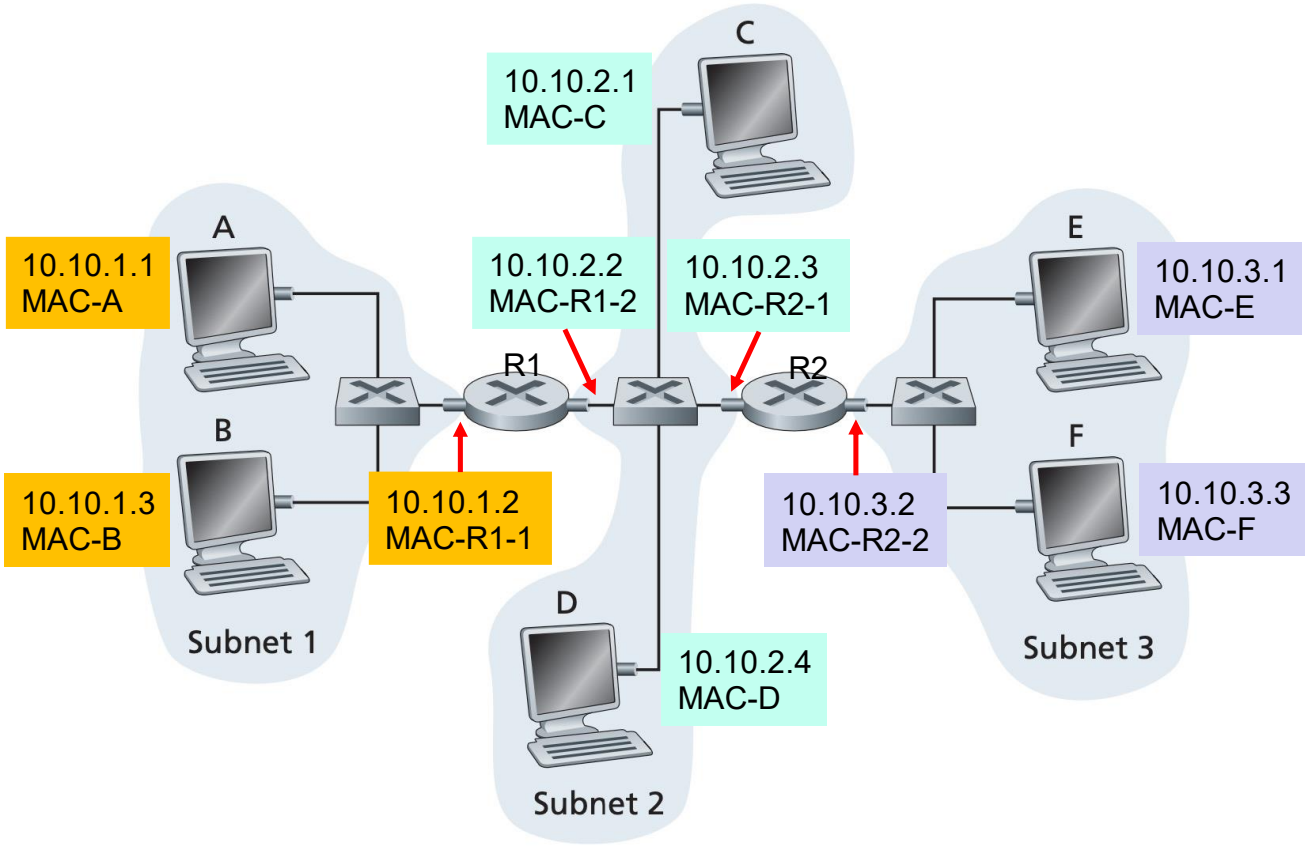
b) 4, D, 0

c) 5, A, 0

Reminder of lecture contents

- Lecture D1-D2
  - ARP
  - CSMA/CD back-off algorithm

Question: Suppose ARP table of all hosts are up to date. What Host A must do to send an IP datagram to Host F.

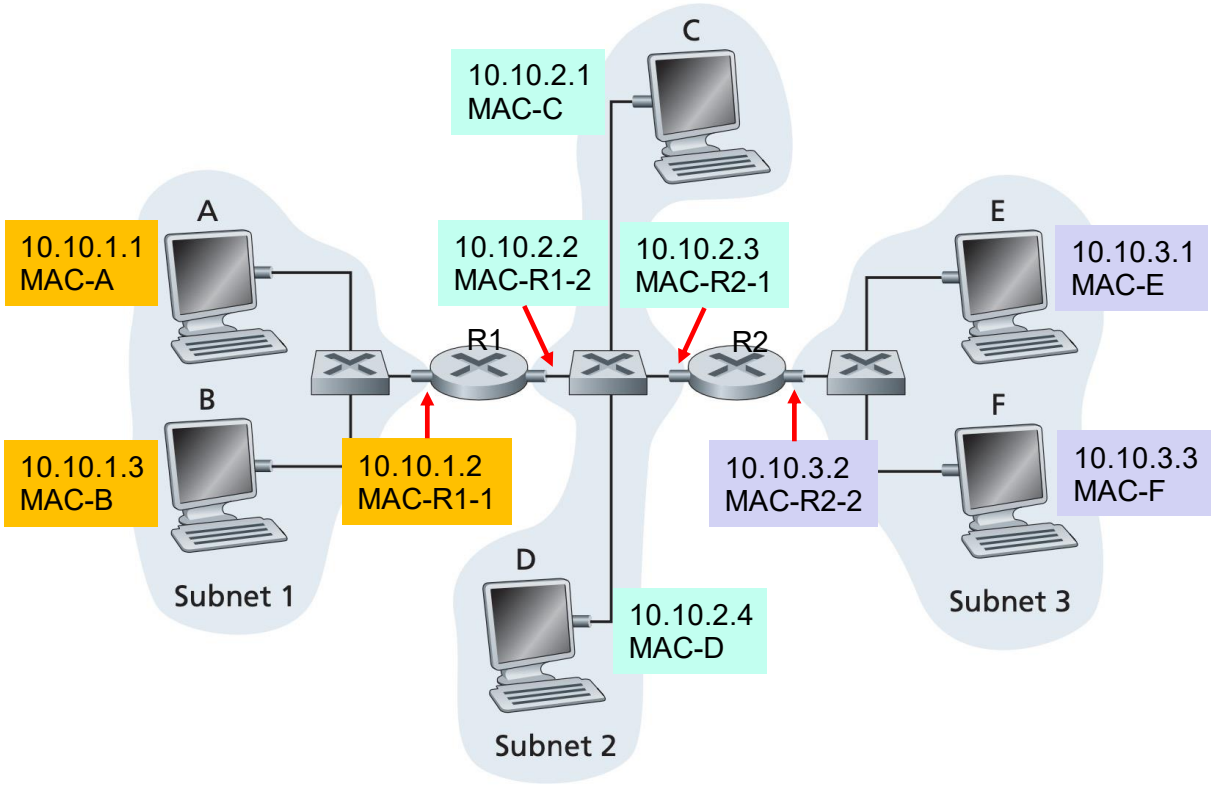


Question: Suppose all of the ARP tables are up to date. What Host A must do to send an IP datagram to Host F.

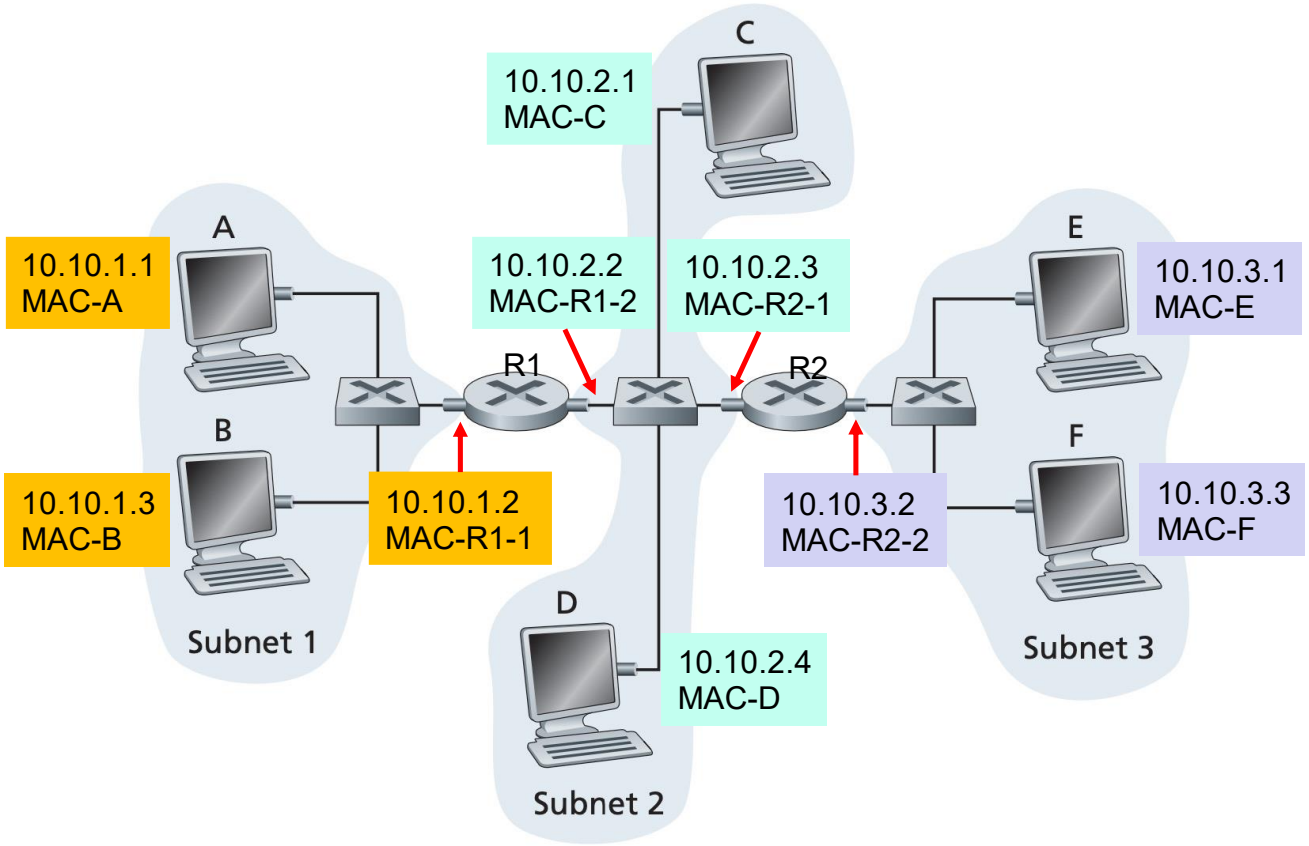
Answer: Host A create IP datagram containing its source IP address (10.10.1.1) and the E's destination IP (10.10.3.3).

Forwarding table in A determines that the datagram should be routed to 10.10.1.2 (R1 gateway IP).

The adapter in A sends a frame (Ethernet packet) containing the R1's destination MAC address (MAC-R1-1) and its source MAC address (MAC-A) via its interface IP (10.10.1.1).



Question: What R1 would do when receiving a frame at interface (MAC-R1-1)?

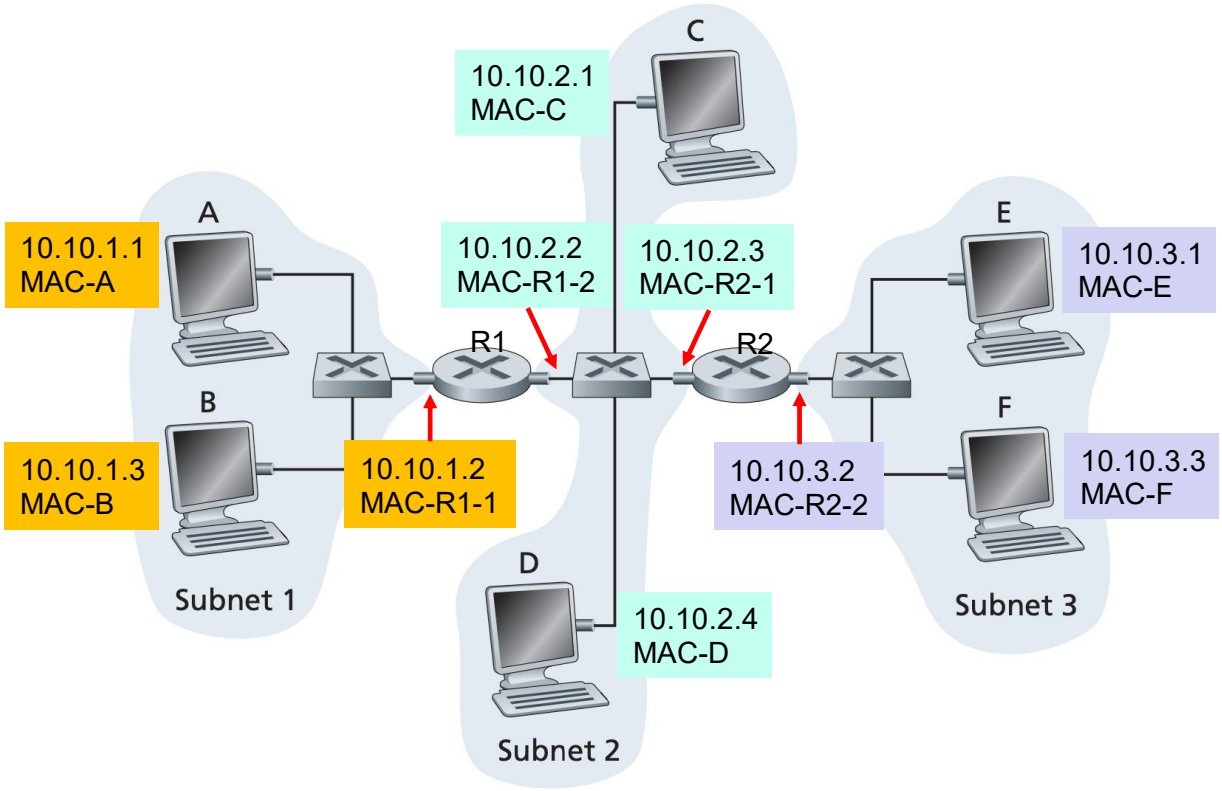


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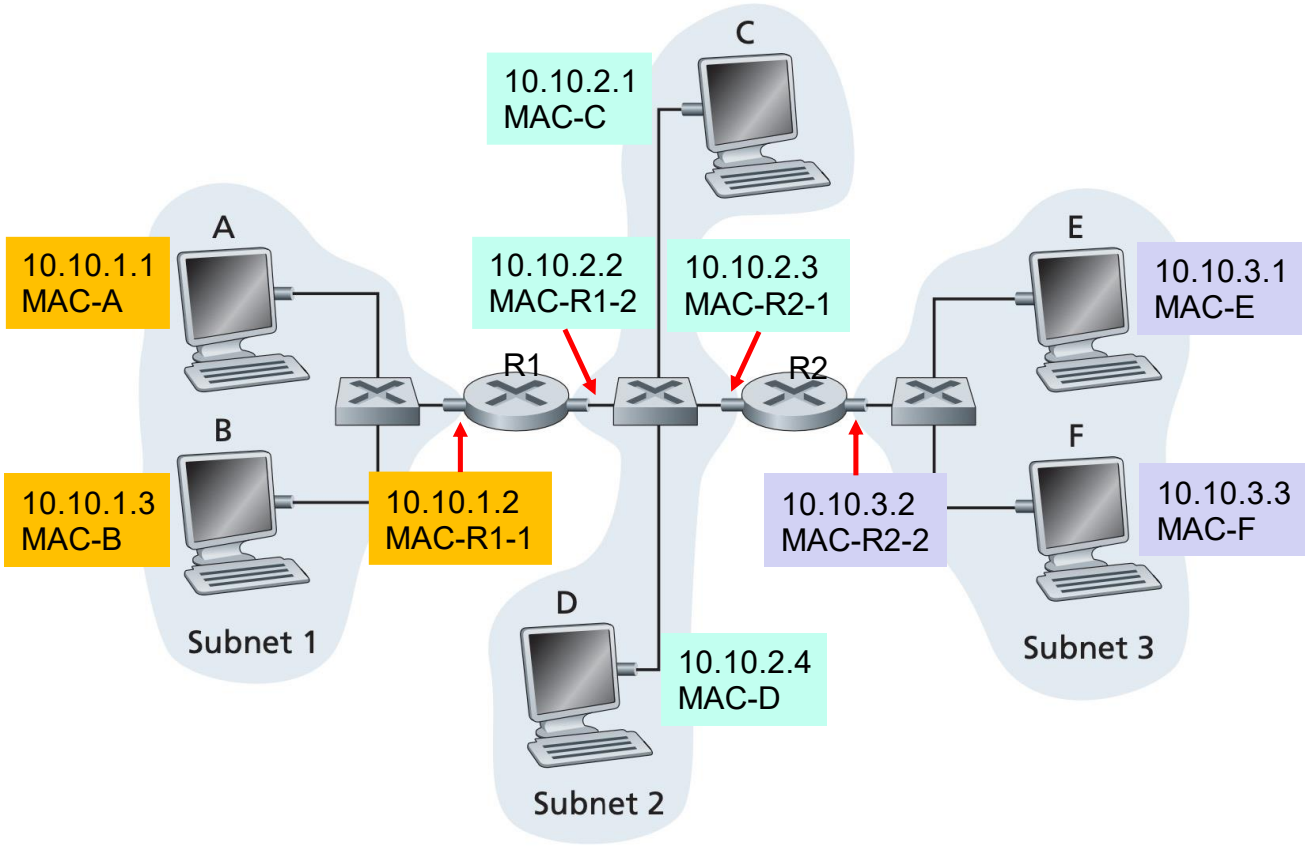
Answer: R1 check the incoming frame if the frame's destination MAC address is matching its own MAC address (MAC-R1-1), and extract the IP datagram.

Forwarding table in R1 determines that the datagram should be routed to 10.10.2.3.

The adapter in R1 sends a frame (Ethernet packet) containing the R2's destination MAC address (MAC-R2-1) and its source MAC address (MAC-R1-2) via its interface with IP address (10.10.2.2)



Question: What R2 would do when receiving a frame at interface (MAC-R2-1)?

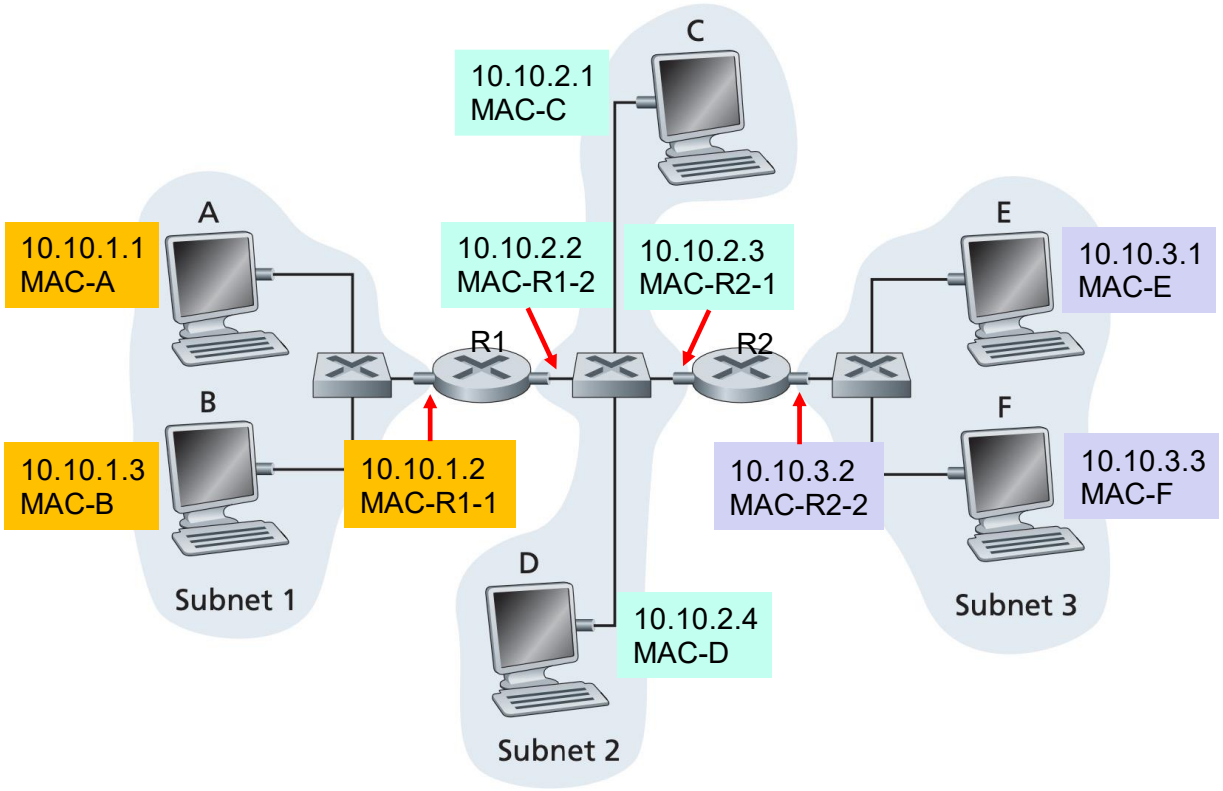


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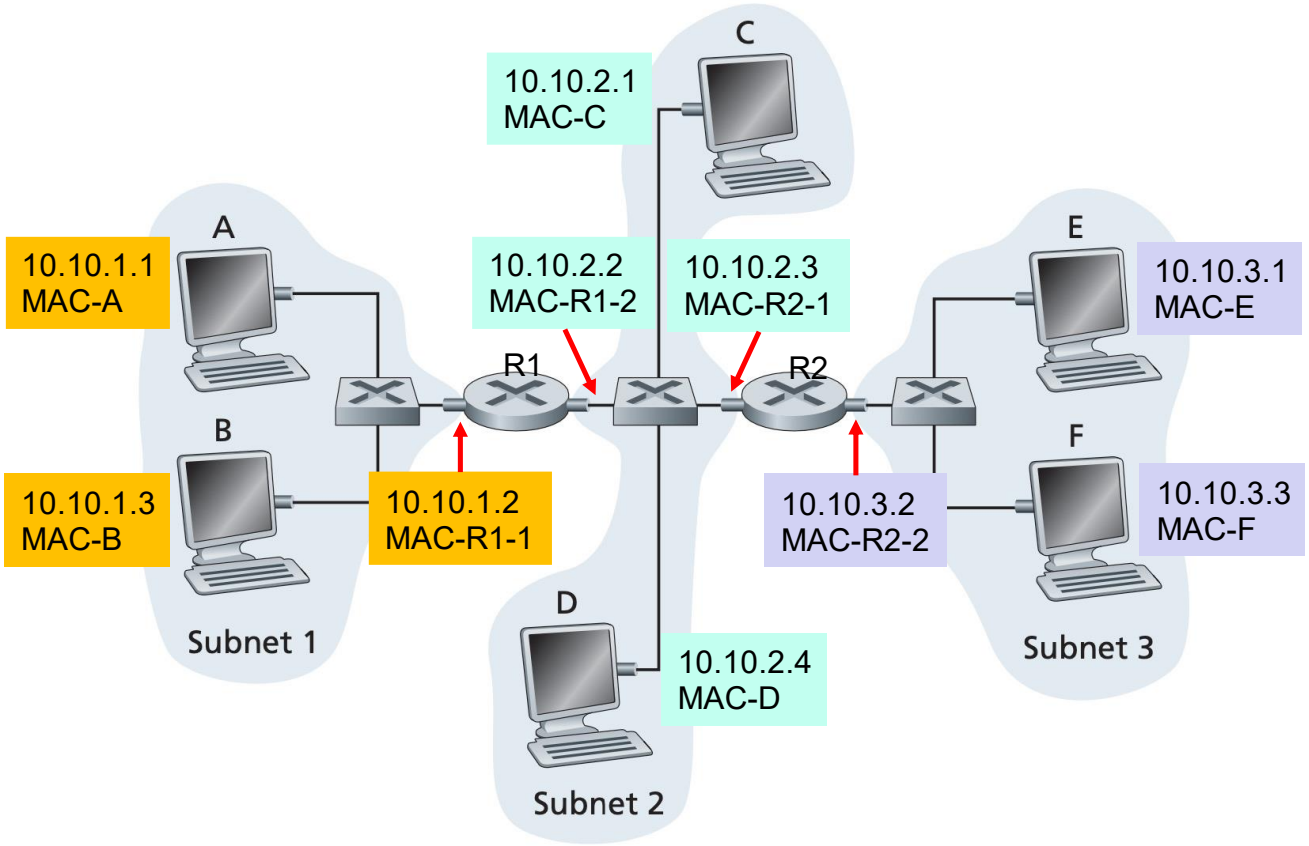
Answer: R2 check the incoming frame if the frame's destination MAC address is matching its own MAC address (MAC-R2-1), and extract the IP datagram.

Forwarding table in R1 determines that the datagram should be routed to 10.10.3.3 (Host F).

The adapter in R2 sends a frame (Ethernet packet) containing the Host F's destination MAC address (MAC-F) and its source MAC address (MAC-R2-2) via its interface with IP address (10.10.3.2)



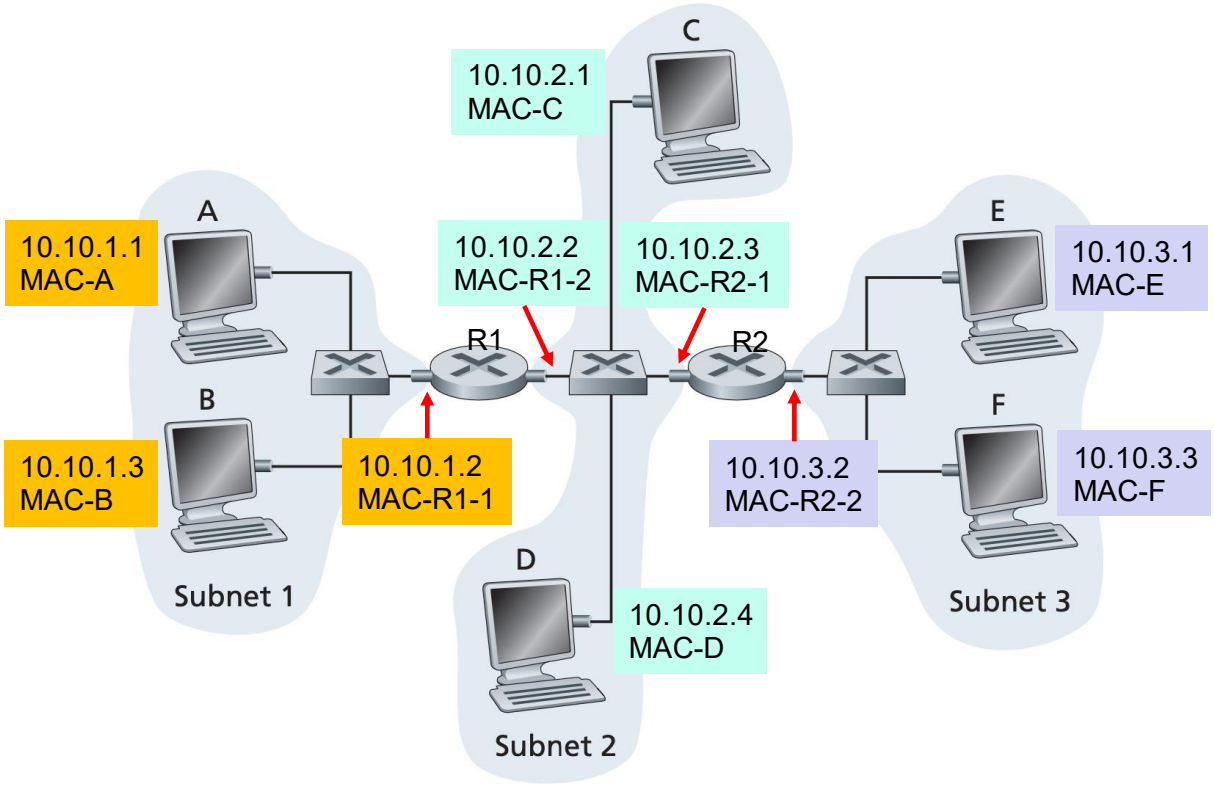
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Answer: Host A must now determine the MAC address of R1 gateway interface IP (10.10.1.2). A sends out an ARP query packet within a broadcast Ethernet frame to MAC address (FF-FF-FF-FF-FF-FF)

R1 receives the query packet and sends to Host A an ARP response packet. This ARP response packet is carried by a frame with MAC destination address (MAC-R1-1).



Ethernet CSMA/CD Back-off algorithm

After aborting the transmission, NIC applies *binary (exponential) backoff* time.

- after *m*th collision, NIC chooses *K* at random from $\{0,1,2, \ldots, 2^m-1\}$. NIC waits K·512 bit times, returns to Step 2
- longer backoff interval with more collisions

Question: A terminal with CSMA/CD detected 3th collision, what is the probability that the back-off algorithm choosing K = 6?



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Answer:

With 3th collision detected, the m=3. Therefore,
K value is in the range between 0 and 7.
The probably of K=6 is 1/8.



Question: With the back-off algorithm choosing K = 6, what is the delay (in seconds) on a 10 Mbps Ethernet?



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Answer:

K·512 = 6 x 512 = 3072 bits

Delay (in seconds) = 3072 bits / 10 Mbits * 1 second
= 307.2 microsecond



Question: CSMA/CD network has a bitrate of 10Mbps and the maximum one-way propagation delay is 2ms. What is the minimum collision window in the form of frame size?



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Answer:

Minimum collision window is equal to end-to-end propagation delay that is 2 * 2ms = 4ms.

Delay (in seconds) = 4 ms / 1s * 10Mbits
= 40 Kbits
= 5 Kbytes

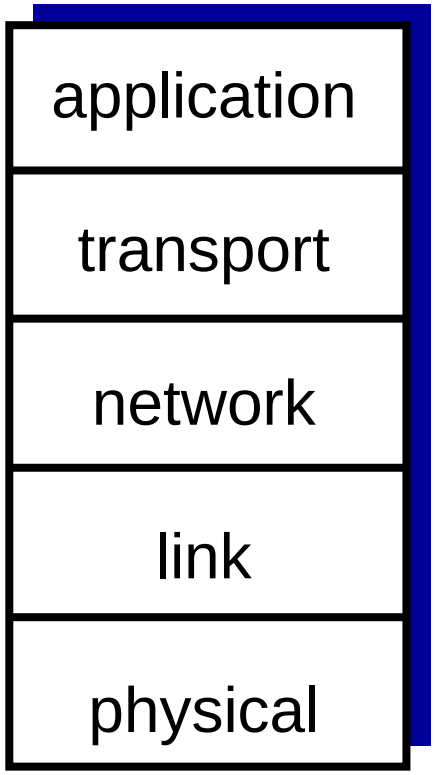


Reminder of lecture contents

- Lecture (Part A)
  - Network basics bits and bytes
  - TCP/IP model

TCP/IP model (internet model)

- *application:* supporting network applications
  - FTP, SMTP, HTTP
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.111 (WiFi), PPP
- *physical:* bits "on the wire"



TCP/IP model (internet model)

Question: A laptop device normally implemented how many layers of OSI model?



TCP/IP model (internet model)

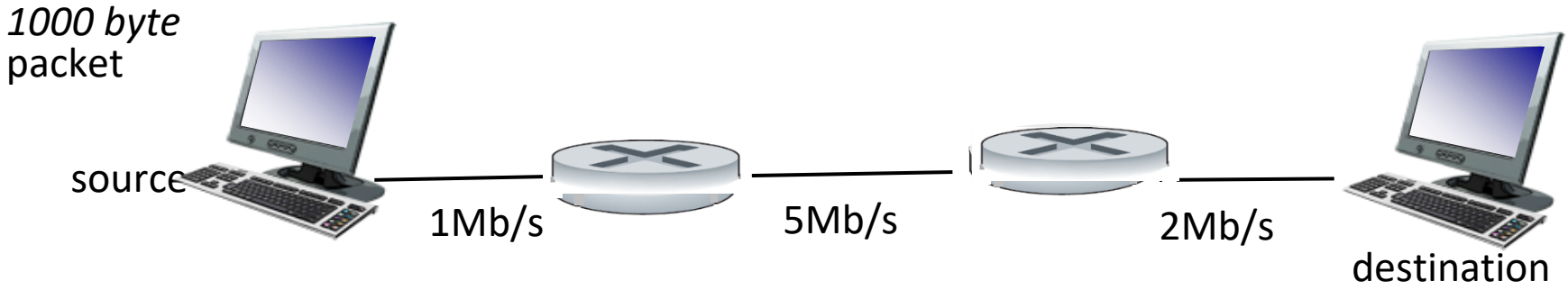
Question: A laptop device normally implemented how many layers of OSI model?

Answer:

- Application
- Transport
- Network
- Data Link
- Physical

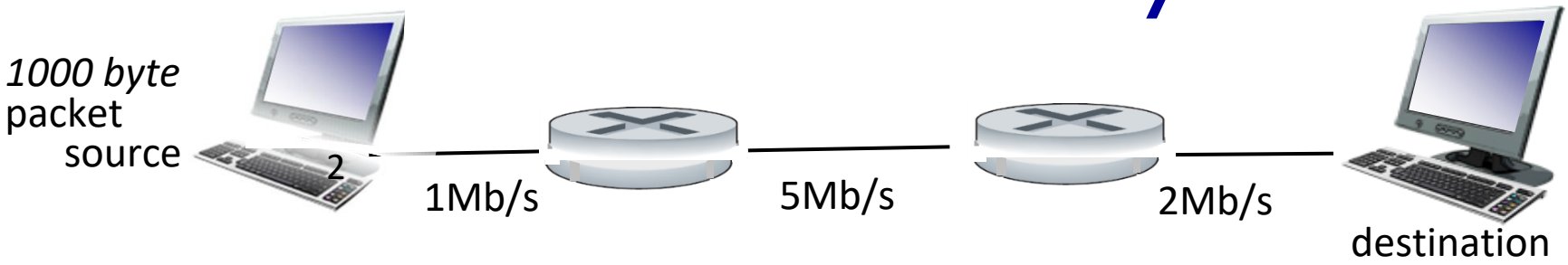


Network basics bits and bytes



- A 1000 byte packet is sent from source to destination through 3 links 1Mb/s, 5Mb/s, 2Mb/s
- Counting only transmission delay:
  - a) What is the end-to-end delay?
  - b) What is the round-trip-time (RTT)?
  - c) How long will it be between starting sending until the 3rd packet reaches the destination (assume 2nd and 3rd packets transmit as soon as they possibly can)?

Network basics bits and bytes



- 1000 bytes = 8000 bits.
- 1Mb/s = 1000000 bits/second
- Delay to first router = 8000/1000000secs= 0.008sec = 8 ms
- Delay to first to second router = 1.6ms
- Delay second to third router = 4ms

a) End-to-end delay = 8+1.6+4ms = 13.6 ms

b) RTT = 2x end-to-end = 27.2ms

- Second packet starts transmission 8ms after first.
- Third packet starts transmission 8x2ms = 16 ms after first.

c) Third packet arrives after 13.6+8x2ms = 29.6ms.

Reminder of lecture contents

- Part B
  - Pipeline protocol
  - TCP round trip time

Pipelined protocols: overview

Go-back-N:

- sender can have up to N unacked packets in pipeline
- receiver only sends *cumulative ACK*
  - doesn't ACK packet if there's a gap
  - Doesn't buffer out of order
- sender has timer for oldest packet
  - when timer expires, retransmit *all* unacked packets

Selective Repeat:

- sender can have up to N packets which it has not seen ACK for in "pipeline"
- Receiver sends *individual ACK* for each packet
- sender maintains timer for each packet with no ACK
  - when timer expires, retransmit only that unacked packet

Pipelined protocols: overview

Question: Which of the following statement is correct :

a) Go-Back-N is more bandwidth efficient because it would retransmit an entire window even if a single packet is lost.

b) Go-Back-N uses cumulative acknowledgements which can reduce the traffic.

c) Selective Repeat is more bandwidth efficient compared to Go-Back-N because it resend only the defective or missing packets and not the entire windows.

d) In Go-Back-N, if a sent packet is found damaged, then all the packets are retransmitted till the last packet.



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d) In Go-Back-N, if a sent packet is found damaged, then all the packets are retransmitted till the last packet.



TCP round trip time, timeout

Q: how to set TCP timeout value?

- longer than RTT
  - but RTT varies
- *too short:* early timeout, unnecessary retransmissions
- *too long:* slow reaction to segment loss

Q: how to estimate RTT?

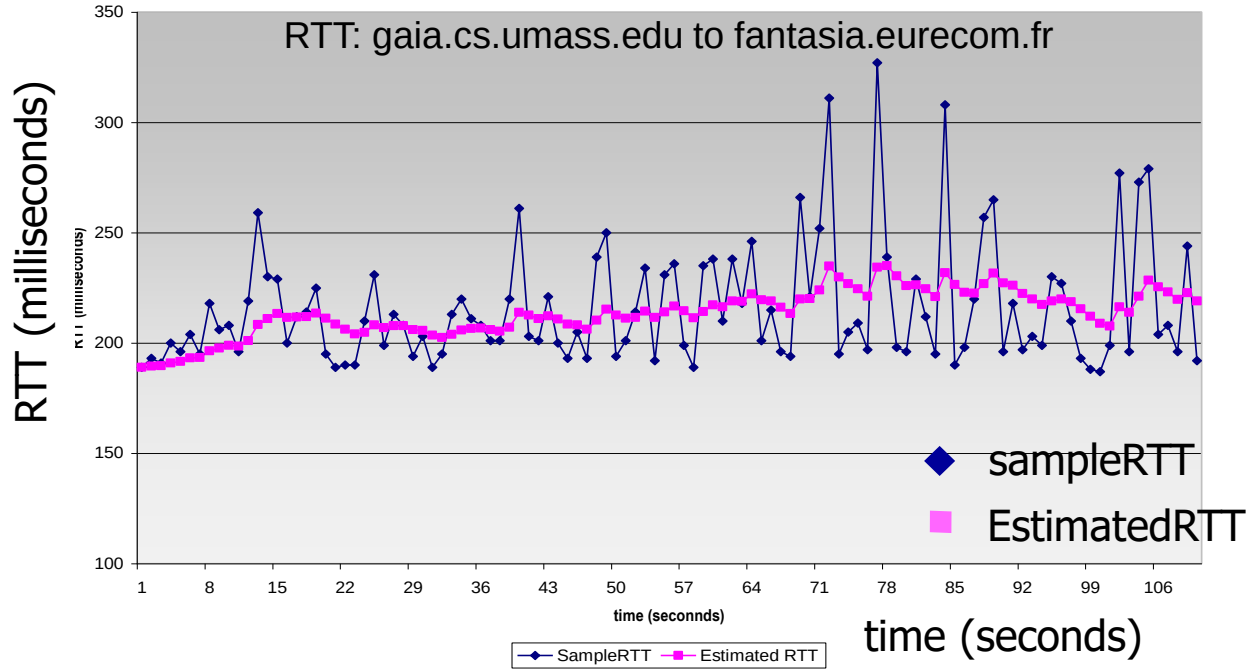
- **`SampleRTT`**: measured time from segment transmission until ACK receipt
  - ignore retransmissions
- **`SampleRTT`** will vary, want estimated RTT "smoother"
  - average several *recent* measurements, not just current **`SampleRTT`**

TCP round trip time, timeout

`EstimatedRTT = (1- α)*EstimatedRTT + α*SampleRTT`

- exponential weighted moving average
- influence of past sample decreases exponentially fast
- typical value: $\alpha = 0.125$

EstimatedRTT_new = 0.875 · EstimatedRTT + 0.125 · SampleRTT



Test your understanding

- What is the problem if a TCP timeout is set too short?
- What is the problem if a TCP timeout is set too long?
- Estimated RTT is 12ms $\alpha = 0.125$, Sample RTT is 8ms. Calculate a new Estimated RTT

Test your understanding

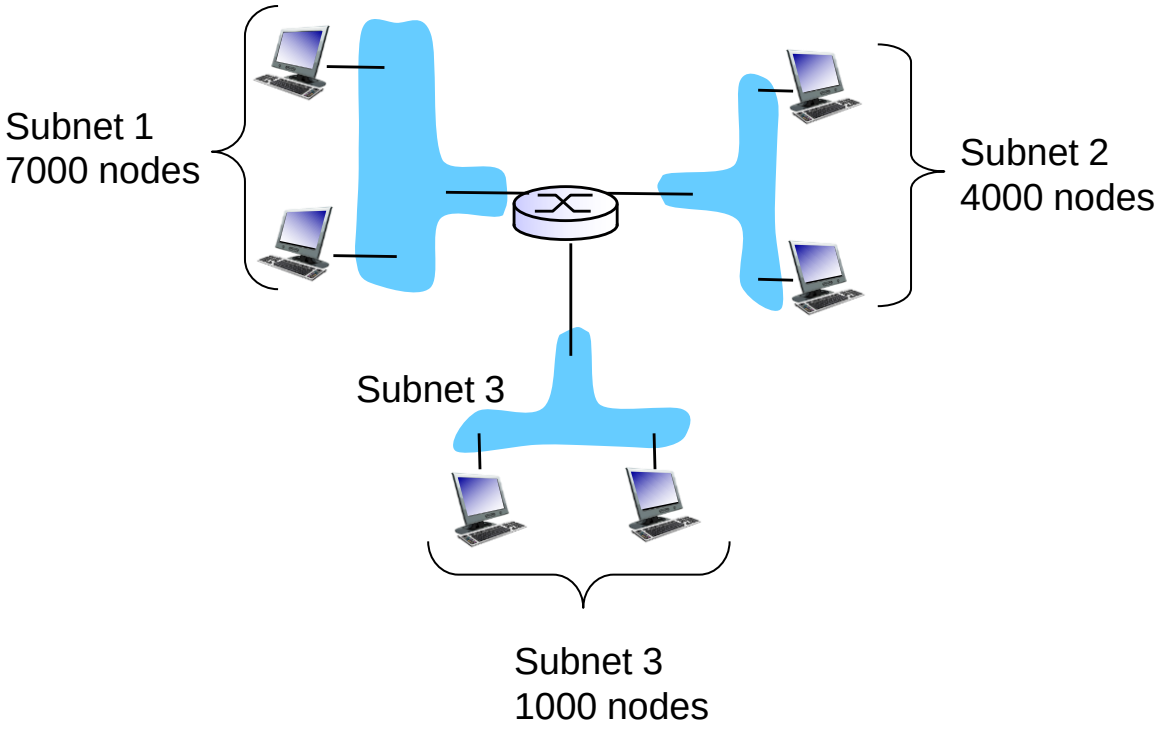
- What is the problem if a TCP timeout is set too short?
  - Packets are retransmitted needlessly (they were delayed not lost) wasting bandwidth
- What is the problem if a TCP timeout is set too long?
  - Packets take a long time to retransmit meaning your connection is slow.
- Estimated RTT is 12ms α = 0.125, Sample RTT is 8ms. Calculate a new Estimated RTT
  - Estimated RTT= (1- α) 12 + α 8 ms = (0.875).12 + 0.125 . 8 ms = 10.5+1ms = 11.5 ms

If you did not get it right revise this section

IP Subnet

Question: Given the IP address of 192.0.0.0/18 which is enough for 16382 nodes, you are required to divide the IP address into 3 subnets, from largest subnet to the smallest.

What is Subnet 1 IP address range?



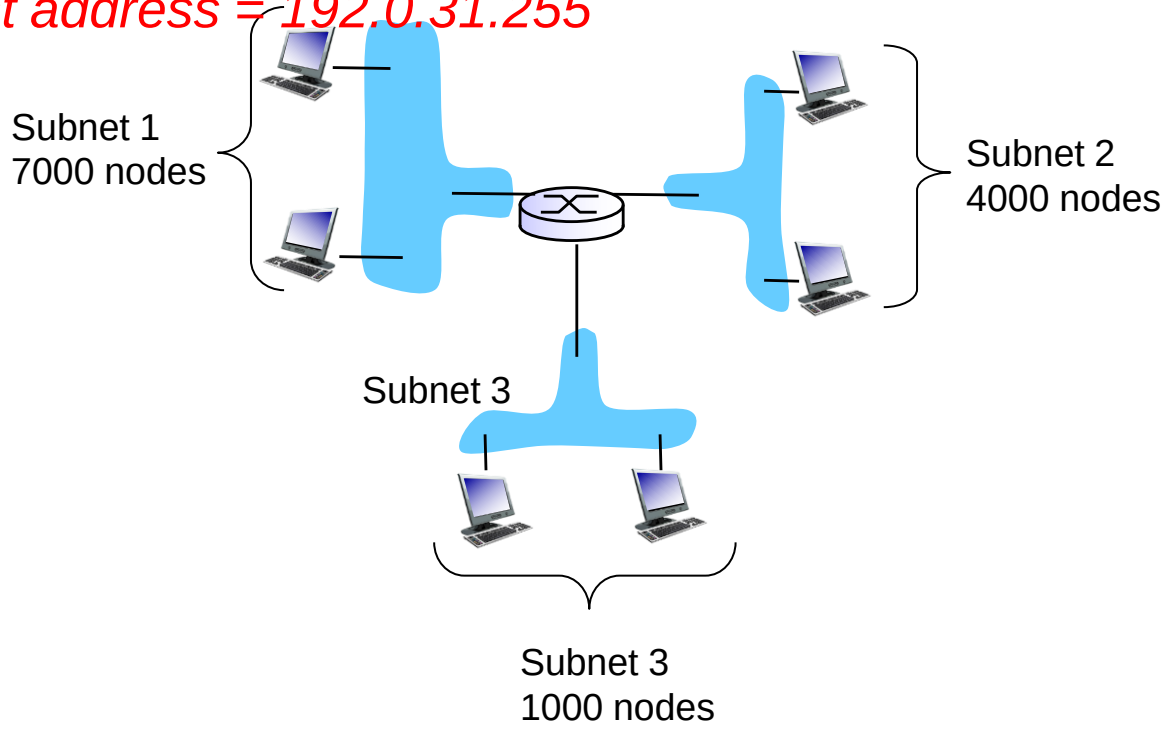
IP Subnet

Answer: We start from 192.0.0.0 with a prefix /19 that creates 8192 addresses enough for all Subnet 1 (7000 nodes), hence the answer are:

(i) IP range of Subnet 1 = 192.0.0.0 – 192.0.31.255

Usable IP range of Subnet 1 = 192.0.0.1 – 192.0.31.254

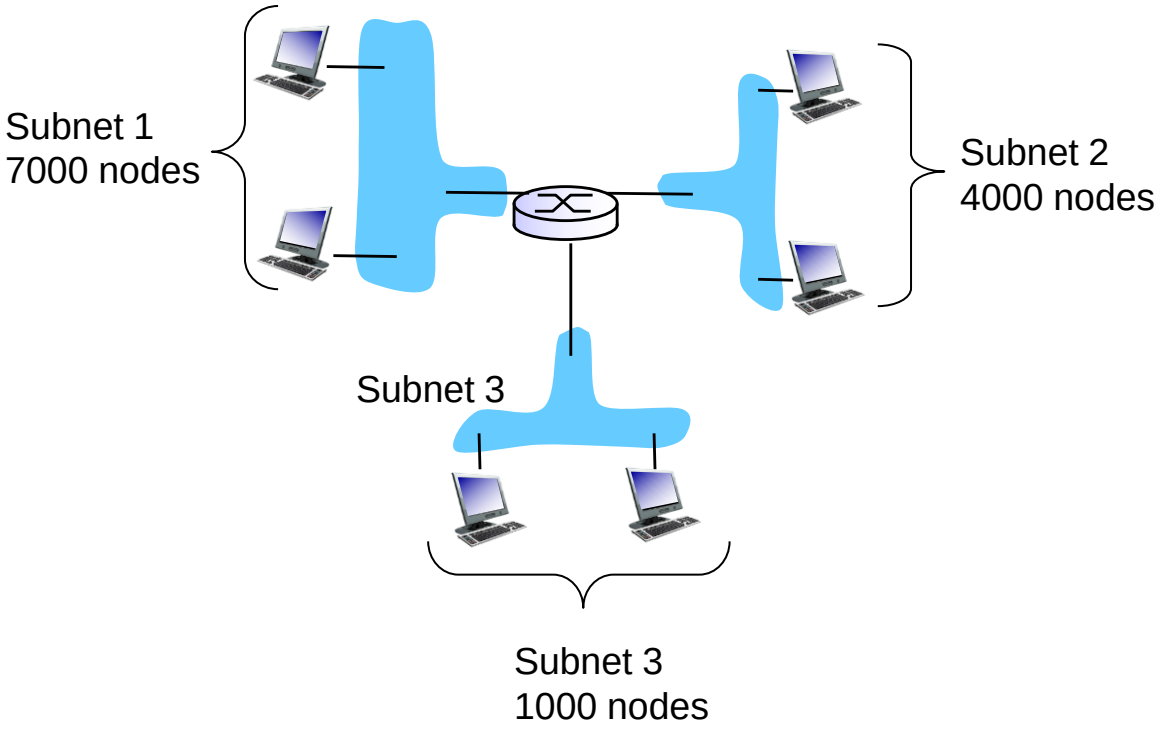
Broadcast address = 192.0.31.255



IP Subnet

Question: Given the IP address of 192.0.0.0/18 which is enough for 16382 nodes, you are required to divide the IP address into 3 subnets, from largest subnet to the smallest.

What is Subnet 2 IP address range?



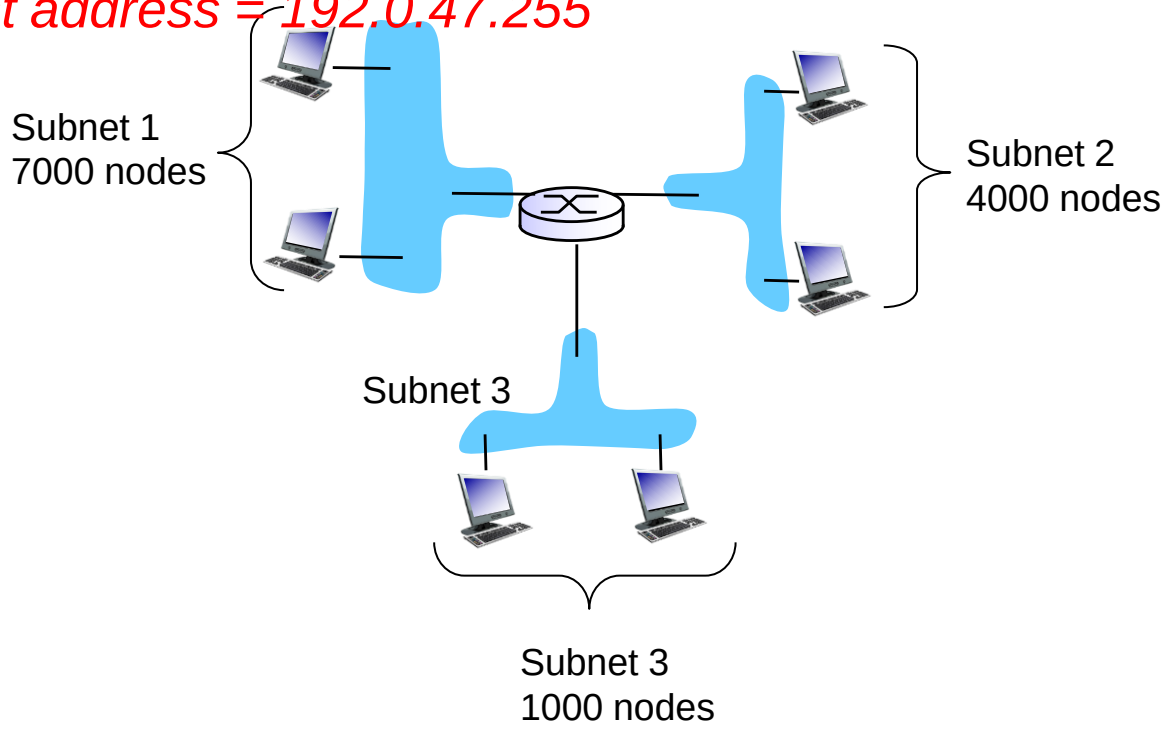
IP Subnet

Answer: We start from 192.0.0.0 with a prefix /19 that creates 8192 addresses enough for all Subnet 1 (7000 nodes), hence the answer are:

(i) IP range of Subnet 1 = 192.0.32.0 – 192.0.47.255

Usable IP range of Subnet 1 = 192.0.32.1 – 192.0.47.254

Broadcast address = 192.0.47.255



Reminder of lecture contents

- Lecture (Part C)
  - IPv6 motivation
  - Dijstra

IPv6: motivation

Question: Which of the following is correct about the IPv6?

a) 32-bit address space soon to be completely allocated.
b) IPv6 header format requires less bandwidth than IPv4
c) IPv6 datagram allows checksum
d) Changes in header format to help speeding up processing/forwarding
e) IPv6 has many overlays to handle Internet growth
f) Changes in header format to facilitate QoS
g) IPv6 datagram allows fragmentation

IPv6: motivation

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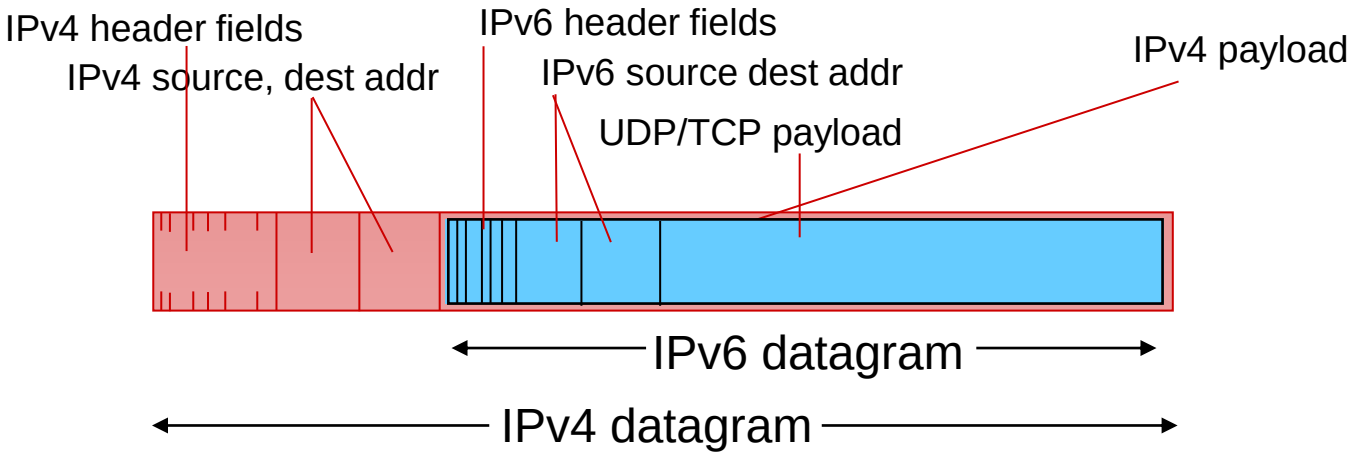
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g) IPv6 datagram allows fragmentation

Transition from IPv4 to IPv6

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- TRUE or FALSE?

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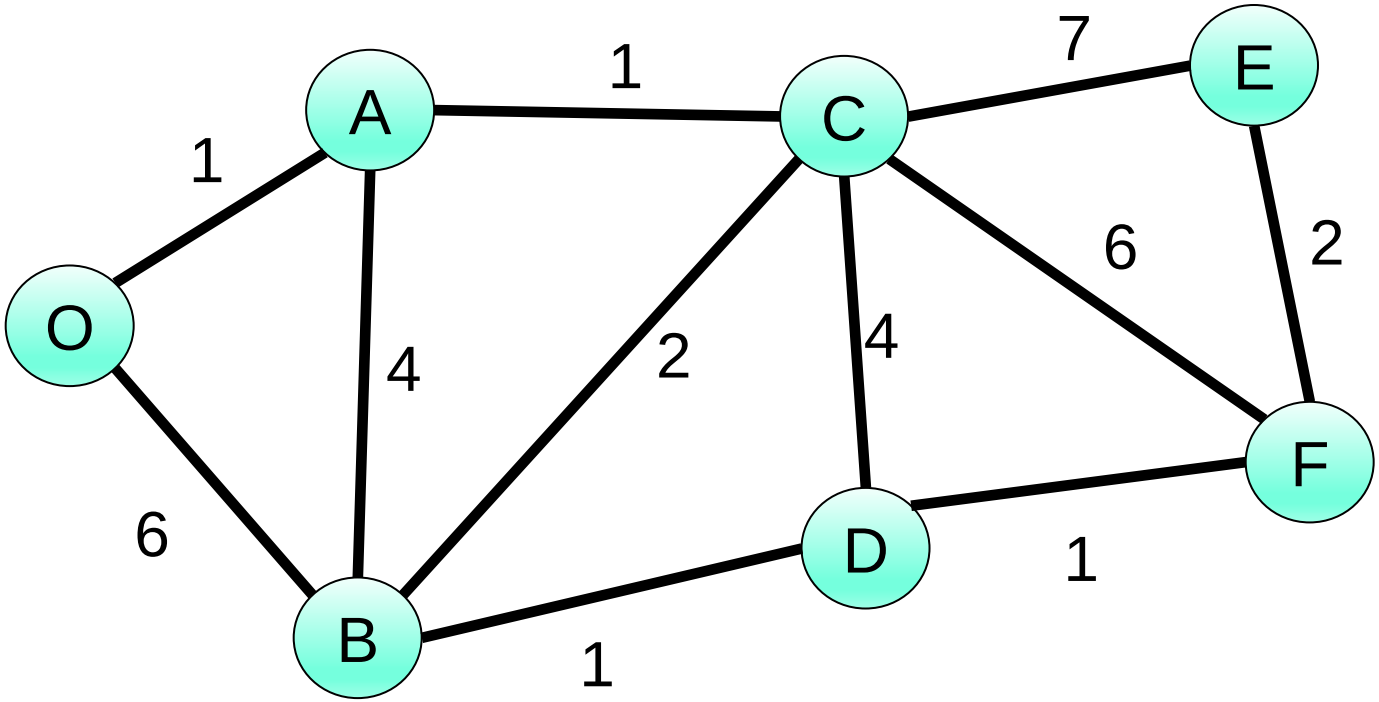


Dijkstra revision

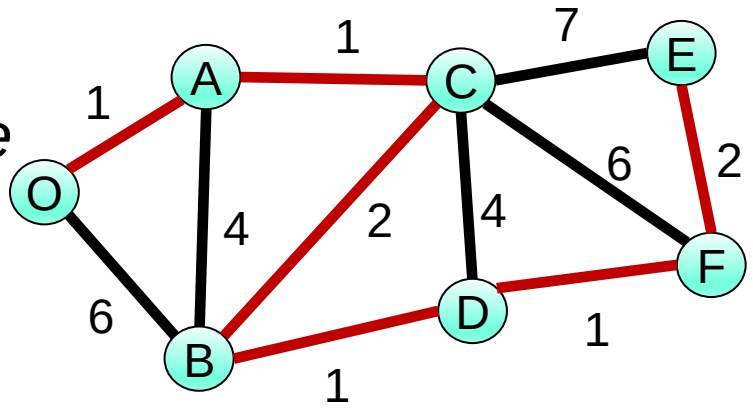
- Key things to remember about Dijkstra
- Need to keep track of:
  - Permanent set of nodes N'= {…}
  - Estimated cost to all nodes D(v)
  - Predecessor node p(v)
- Initialise (imagine we start from node A)
  - N'={A}
  - If node connects to A D(v) = d(A,v)
  - If node connects to A p(v) = A
- At every step – find node not in N' that has smallest D(v) --- call this node B for now:
  - N':= N'UB (add B into N')
  - Update distance to any node that has a connection to B
  - For all v not in N' D(v) = min(D(v), D(B) + c(B,v))

Dijkstra Example network

- Find shortest path to all destination nodes. starting from O.



Greedily pick the closest node at every stage



N'	D(A)p(A)	D(B)p(B)	D(C)p(C)	D(D)p(D)	D(E)p(E)	D(F)p(F)
O	1,O	6,O	-	-	-	-
O,A	1,O	5,A	2,A	-	-	-
O,A,C	1,O	4,C	2,A	6,C	9,C	8,C
OACB	1,O	4,C	2,A	5,B	9,C	8,C
OACBD	1,O	4,C	2,A	5,B	9,C	6,D
OACBDF	1,O	4,C	2,A	5,B	8,F	6,D
OACBDFE	1,O	4,C	2,A	5,B	8,F	6,D